

## K-Medoid Clustering – Solved Example – 1

### Step 1

- Select two medoids
- $C1 = (3, 4)$
- $C2 = (7, 4)$
- $Manhattan\ Dist = |x_1 - x_2| + |y_1 - y_2|$
- $Mdist[(2, 6), (3, 4)] = |2 - 3| + |6 - 4| = 3$
- $Mdist[(3, 4), (3, 4)] = |3 - 3| + |4 - 4| = 0$

| i   | x | y | C1 | C2 | Cluster |
|-----|---|---|----|----|---------|
| X1  | 2 | 6 | 3  | 7  |         |
| X2  | 3 | 4 | 0  | 4  |         |
| X3  | 3 | 8 | 4  | 8  |         |
| X4  | 4 | 7 | 4  | 6  |         |
| X5  | 6 | 2 | 5  | 3  |         |
| X6  | 6 | 4 | 3  | 1  |         |
| X7  | 7 | 3 | 5  | 1  |         |
| X8  | 7 | 4 | 4  | 0  |         |
| X9  | 8 | 5 | 6  | 2  |         |
| X10 | 7 | 6 | 6  | 2  |         |

## K-Medoid Clustering – Solved Example – 1

- $C1: \{(2, 6), (3, 4), (3, 8), (4, 7)\}$
- $C2: \{(6, 2), (6, 4), (7, 3), (7, 4), (8, 5), (7, 6)\}$

## K-Medoid Clustering – Solved Example – 1

- $C1: \{(2, 6), (3, 4), (3, 8), (4, 7)\}$
- $C2: \{(6, 2), (6, 4), (7, 3), (7, 4), (8, 5), (7, 6)\}$
- Calculate the Total Cost
- $Cost(c, x) = \sum_i |c_i - x_i|$
- $Total\ Cost = \{Cost((3, 4), (2, 6)) + Cost((3, 4), (3, 8)) + Cost((3, 4), (4, 7)) + Cost((7, 4), (6, 2)) + Cost((7, 4), (6, 4)) + Cost((7, 4), (7, 3)) + Cost((7, 4), (8, 5)) + Cost((7, 4), (7, 6))\}$

## K-Medoid Clustering – Solved Example – 1

- C1: {(2,6), (3,4), (3,8), (4,7)}
- C2: {(6, 2), (6, 4), (7, 3), (7, 4), (8, 5), (7,6)}
- Calculate the Total Cost
- $Cost(c, x) = \sum_i |c_i - x_i|$  1+2      0+4
- $Total Cost = \{Cost((3,4), (2,6)) + Cost((3,4), (3,8)) + Cost((3,4), (4,7)) + Cost((7,4), (6,2)) + Cost((7,4), (6,4)) + Cost((7,4), (7,3)) + Cost((7,4), (8,5)) + Cost((7,4), (7,6))\}$
- $Total Cost = 3 + 4 + 4 + 2 + 3 + 1 + 1 + 2 = 20$  ●

## K-Medoid Clustering – Solved Example – 1

Step 3

- New Medoids
- C1=(3, 4) and O=(7, 3)
- $Manhattan Dist = |x_1 - x_2| + |y_1 - y_2|$
- $Mdist[(2, 6), (7, 3)] = |2 - 7| + |6 - 3| = 8$

| i   | x | y | C1 | O | Cluster |
|-----|---|---|----|---|---------|
| X1  | 2 | 6 | 3  | 8 | C1      |
| X2  | 3 | 4 | 0  | 5 | C1      |
| X3  | 3 | 8 | 4  | 9 | C1      |
| X4  | 4 | 7 | 4  | 7 | C1      |
| X5  | 6 | 2 | 5  | 2 | O       |
| X6  | 6 | 4 | 3  | 2 | O       |
| X7  | 7 | 3 | 5  | 0 | O       |
| X8  | 7 | 4 | 4  | 1 | O       |
| X9  | 8 | 5 | 6  | 3 | O       |
| X10 | 7 | 6 | 6  | 3 | O       |

## K-Medoid Clustering – Solved Example – 1

- C1: {(2,6), (3,4), (3,8), (4,7)}
- O: {(6, 2), (6, 4), (7, 3), (7, 4), (8, 5), (7,6)}
- Calculate the Total Cost
- $Cost(c, x) = \sum_i |c_i - x_i|$
- $Current Total Cost = \{Cost((3,4), (2,6)) + Cost((3,4), (3,8)) + Cost((3,4), (4,7)) + Cost((7,3), (6,2)) + Cost((7,3), (6,4)) + Cost((7,3), (7,4)) + Cost((7,3), (8,5)) + Cost((7,3), (7,6))\}$
- $Current Total Cost = 3 + 4 + 4 + 2 + 2 + 1 + 3 + 3 = 22$  ●

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Step 4

- Cost of Swapping of medoid C2 with O ✓
- $S = \text{Current Total Cost} - \text{Previous Total Cost}$
- $S = 22 - 20 = 2 > 0$
- Hence Swapping C2 with O is not a good Idea.
- Final Medoids are C1=(3, 4) and C2=(7, 4)